LECTURE - 4

Attention is all you need

↓

Transformers

↓

Sequential modeling

↳ CNNs / RNNs → Problem / Issues

→ NOT in parallel

↓

one token at a time

GPT open AI → 2018

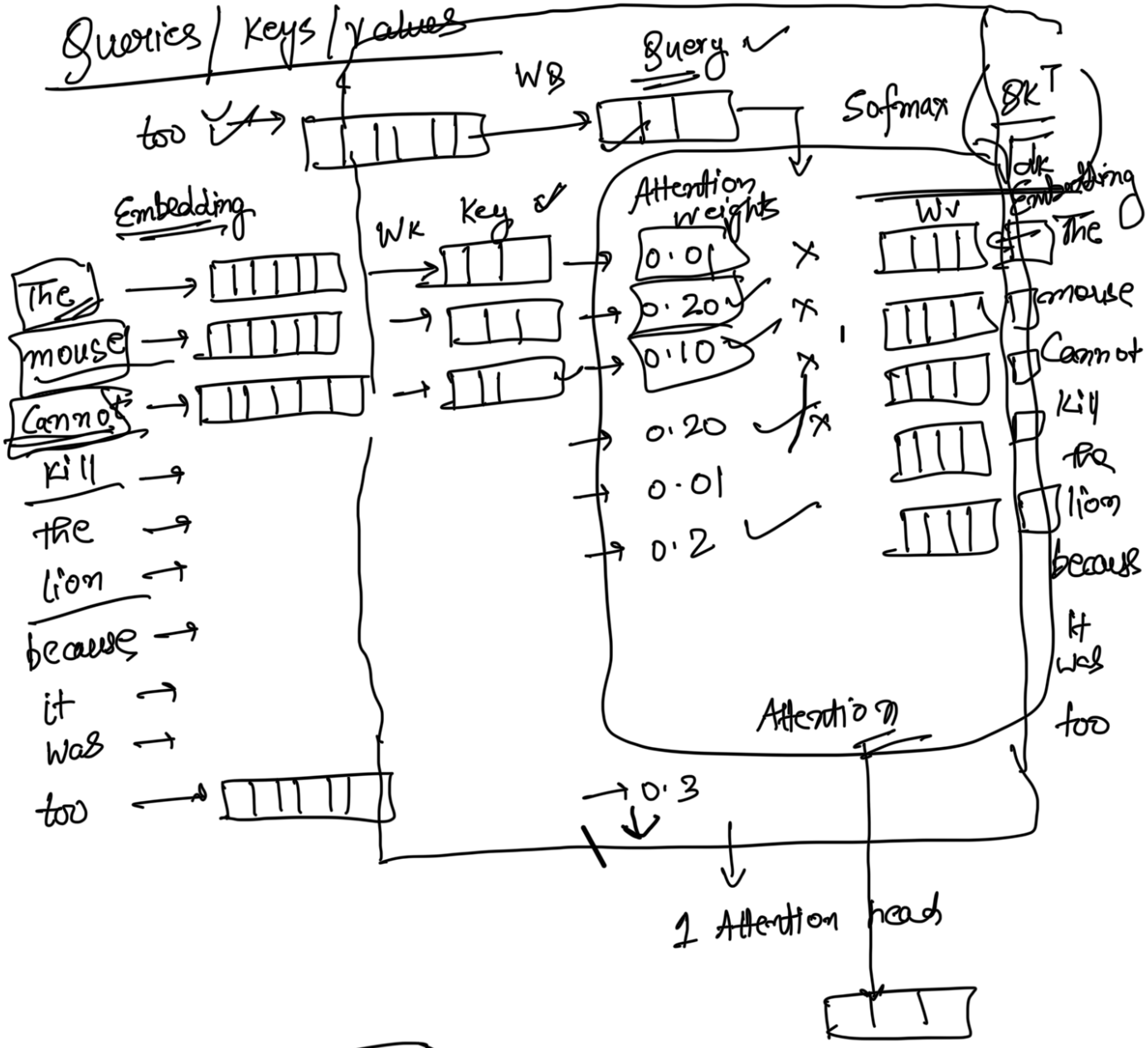
Improving language understanding by Generative Pre-training

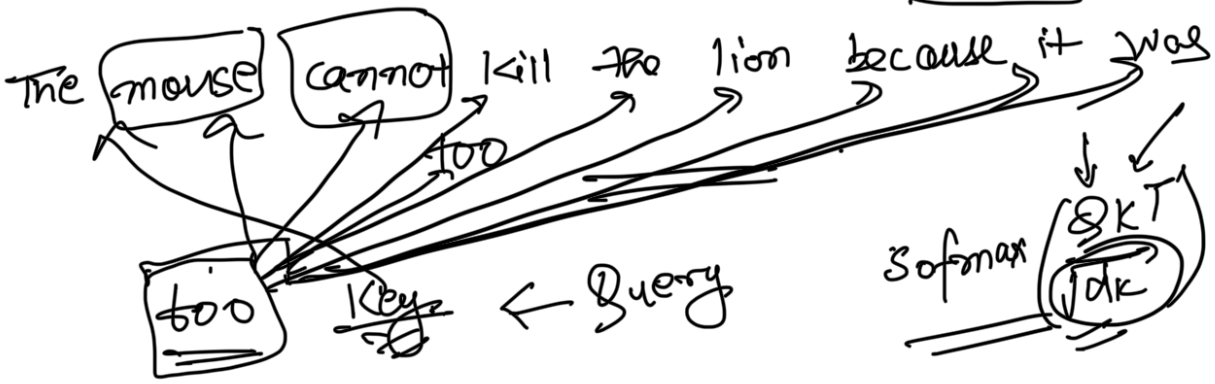
↳ GPT-2 / 3 / 3.4 / 4

Attention

The mouse cannot kill the lion because it was too small

The mouse cannot afraid kill the ~~[illegible]~~ mouse because





QKV

Q → Query → token looking for
K → Key → what offers
V → Value → actual information

$X \rightarrow$ input embedding

$Q = XW_Q \qquad K = XW_K \qquad V = XW_V$

$W_Q W_K V_K$ learned matrix/cos

$QK^T \rightarrow$ High dot product → strong, soft $\left(\frac{QK^T}{\sqrt{d}}\right)V$

low dot product → weak relev

too

$[1 \quad 1 \quad 0]$

The mouse cannot

$[0.01 \quad 00 \quad 0] \rightarrow 0.01$ ← less

$[0.2 \quad 00 \quad 0] \rightarrow 0.2$

$[00 \quad 0.1 \quad 0] \rightarrow 0.1$

d_k

$QK^T \rightarrow$

$$\frac{QK^T}{\sqrt{d_k}}$$

Softmax $\rightarrow$ Softmax $\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$

values

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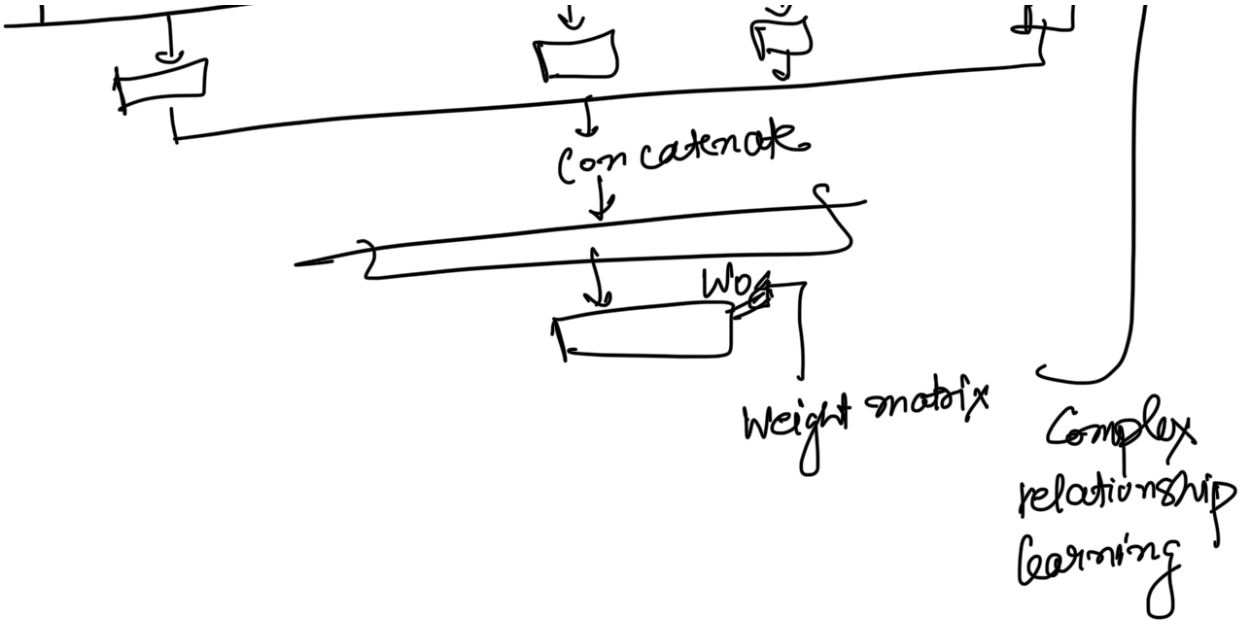
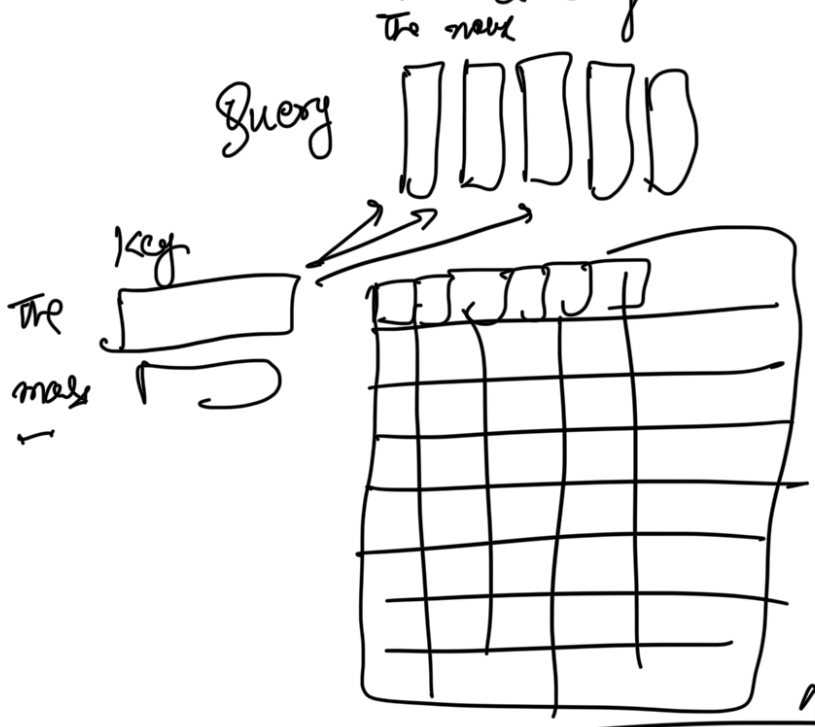
why $\sqrt{d_k}$ and not d_k ?

Multihead Attention

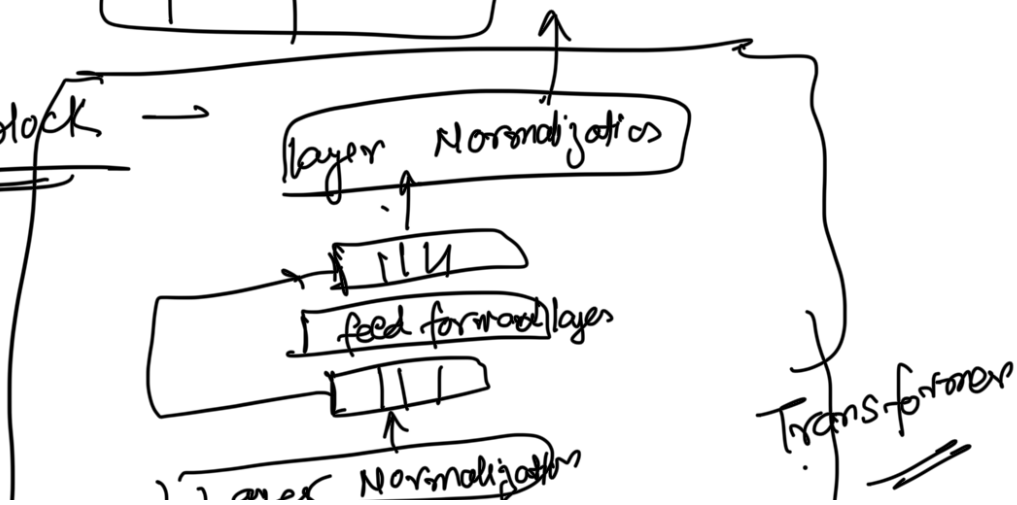
AH Attention head | AH | AH | AH

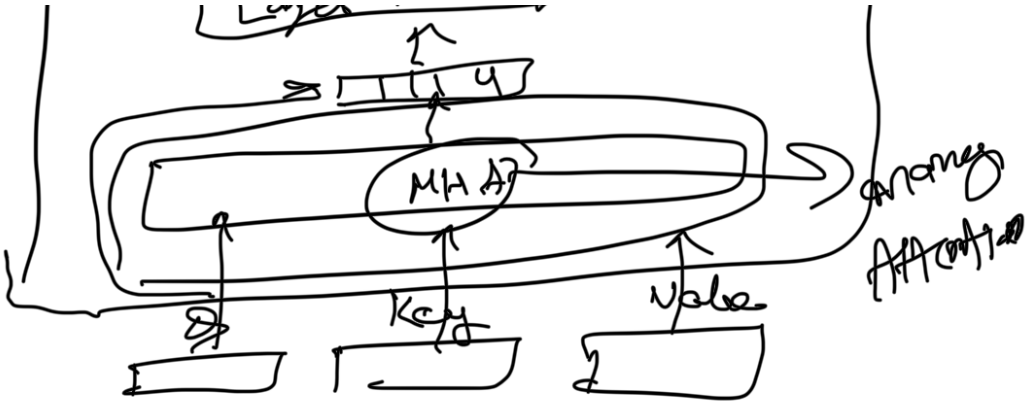


Casual Masking → query to attention head is a single vector.



Transformer block →





The cat chased the mouse was tired
that

Positional Encoding →

The dog looked at the boy and barked
sit down
~~Run away~~

The boy looked at the dog and smiled
touched
patted
scared

Vaibhav → how/which mouse?